The average of the logarithm of the smallest divisor above the square root is asymptotically linear

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Editorial note. The counted statement is the author's unlabelled cor environment at source0.tex lines 248-253; its proof environment opens at line 255, immediately after the blank separator, so the pairing is by adjacency and the retained proof range 255-265 is the whole environment.

Editorial note. The numbered identity eq:dual wasserstein (source lines 243-246) is retained as context because the counted proof refers to it twice, at source lines 260 and 264; it is the only outside label of the retained proof.

Editorial note. Because the corollary carries no author label, its identity is recorded by the pinned source position of the two environments (statement lines 248-253, proof lines 255-265) and by the claim key, which encodes that position.

Editorial note. This article carries no label-number dump and the retained counter values are left empty for the whole source, so the printed numbering of the card is the wrapper's default; the original printed number of this unlabelled corollary is not recoverable from the pinned material and is therefore not asserted.

Editorial scope. One original result of the pinned publisher article, counted once: the asymptotic for the sum of the logarithm of the smallest divisor above the square root (unlabelled corollary at source0.tex lines 248-253).

Allowed standard prerequisites (editorial).

- The function rho. rho(n) is the least divisor of n that is at least the square root of n.. It is the function whose average is estimated.
- The Wasserstein-type identity for Lipschitz statistics. For a bounded Lipschitz function of the normalised logarithmic parts, the expectation of the truncated version equals the expectation of the same function of the Dirichlet limit plus an error of order 1/log x.. It is applied twice, to a specific Lipschitz function and to the divisors themselves.
- Partial summation. Sums of the form sum over n up to x of a quantity divided by log n can be converted into Abel sums.. It passes from the normalised sum to the unnormalised asymptotic.

$$\mathbb{E}\Big[\mathbb{1}_{N_x\geq 2}\cdot\psi\big(\tfrac{\log P_1}{\log N_x},\tfrac{\log P_2}{\log N_x},\dots\big)\Big]=\mathbb{E}\big[\psi\big(V_1,V_2,\dots\big)\big]+O_\psi(1/\log x).\qquad(1)$$

Corollary 0.1. *Let* $\rho(n)\coloneqq\min\{d|n:d\geq\sqrt{n}\}$. *Then there is a constant* $c\in(1/2,1)$ *such that*

$$\sum_{n\leq x}\log\rho(n)=cx\log x+O(x)\qquad(x\geq 1).$$

Proof. Let

$$\psi(v_1,v_2,\dots)\coloneqq\inf\big(\mathcal{D}(v_1,v_2,\dots)\cap[\tfrac{1}{2},1]\big)$$

with $\mathcal{D}(v_1,v_2,\dots)\coloneqq\{\sum_{i\in B}v_i:B\subseteq\mathbb{N}\}$. It is easy to check that this is a 1-Lipschitz function. Note that the left-hand side of (1) with this definition for ψ is exactly equal to

$$\frac{1}{\lfloor x\rfloor}\sum_{2\leq n\leq x}\frac{\log\rho(n)}{\log n}.$$

With (1) and partial summation, we prove the corollary with $c=\mathbb{E}[\psi(\boldsymbol{V})]$. □

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